Tutorial 4

1. Determine the output 'F' of this combinational circuit as shown in fig. 1.

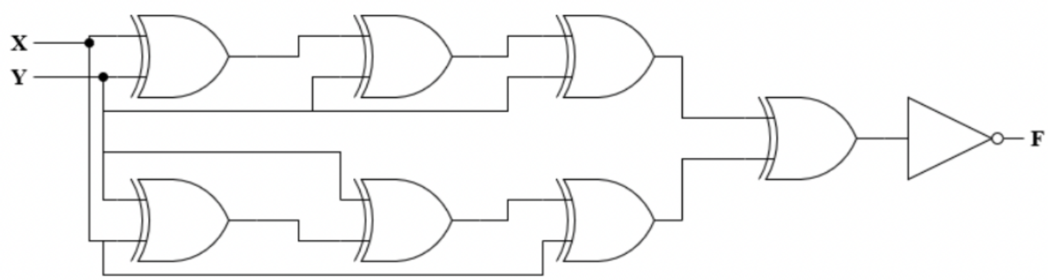


Figure 1: Circuit for Q1

Sol.

$$F = \overline{F_1 \otimes F_2}$$

$$F_1 = X \otimes Y \otimes Y \otimes Y$$

Since $Y \otimes Y = 0$

$$F_1 = X \otimes Y \otimes 0 \qquad (Y \otimes 0 = Y)$$

$$F_1 = \underline{\underline{X \otimes Y}}$$

Similarly $F_2 = X \otimes Y \otimes Y \otimes X \quad = \underline{\underline{0}}$

Now, $F = \overline{F_1 \otimes F_2}$

$$F = \overline{(X \otimes Y) \otimes 0} = \overline{X \otimes Y} = \underline{\underline{X \odot Y}}$$

(XNOR)

2. Implement a combinational logic circuit to convert 4-bit binary input to Gray code.

Sol.

Binary				Gray code			
b_3	b_2	b_1	b_0	g_3	g_2	g_1	g_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

kmap for $\underline{g_0}$

$b_3 b_2 \backslash b_1 b_0$	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$$g_0 = \bar{b}_1 b_0 + b_1 \bar{b}_0 = b_1 \oplus b_0 = b_0 \oplus b_1$$

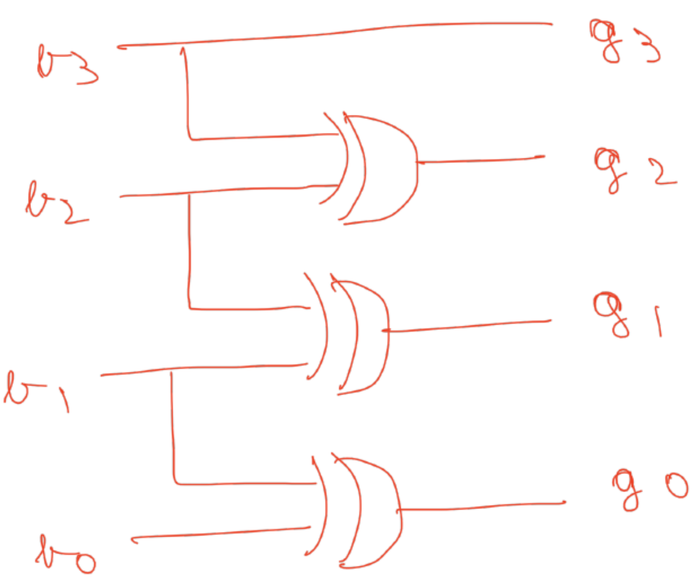
Similarly for all other bits, we get

$$g_1 = b_2 \bar{b}_1 + b_1 \bar{b}_2 = b_1 \oplus b_2$$

$$g_2 = b_2 \bar{b}_3 + b_3 \bar{b}_2 = b_2 \oplus b_3$$

$$g_3 = b_3.$$

Circuit →



3. Design a 3-bit combinational circuit for which, the output is 1 when the number of 1's is less than 0's and 0 otherwise. This circuit can also be called a minority circuit. Design the circuit using suitable logic gates.

Sol.

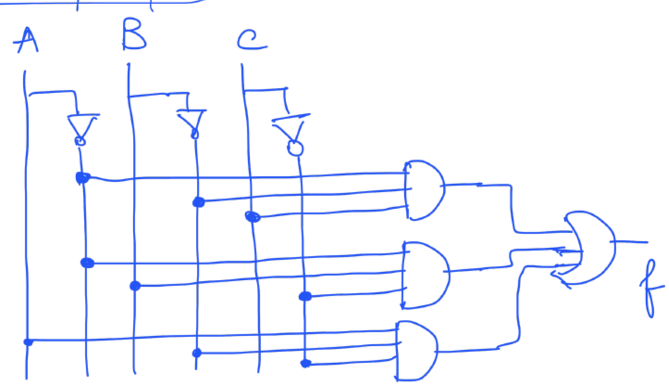
4) Truth table

A	B	C	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

$$f = \Sigma m(1,2,4) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

C \ AB	00	01	11	10
0	0	1: 1	3	2: 1
1	4: 1	5	7	6

Circuit



4. Design a 4-bit combinational circuit specified by the following Boolean function using only NAND and only NOR gates.

$$F(A, B, C, D) = \sum(1, 3, 5, 7, 9, 12) + \sum d(2, 10).$$

Sol.

5) $f(A,B,C,D) = \sum(1,3,5,7,9,12) + \sum d(2,10)$

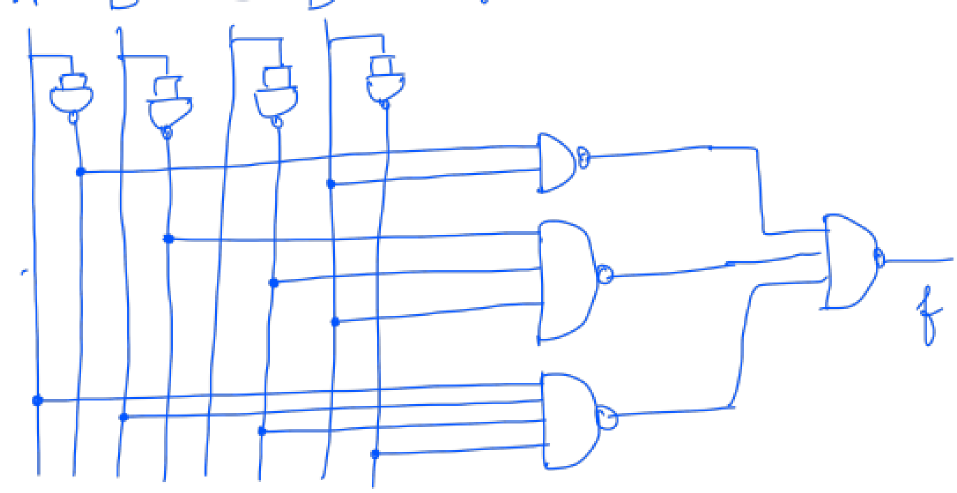
AB \ CD	00	01	11	10
00	0	1: 1	3: 1	2: X
01	4	5: 1	7: 1	6
11	12: 1	13	15	14
10	8	9: 1	11	10: X

$$f = \bar{A}D + \bar{B}\bar{C}D + AB\bar{C}\bar{D}$$

NAND implementation

$f = \bar{A}D + \bar{B}\bar{C}D + AB\bar{C}\bar{D}$

A B C D



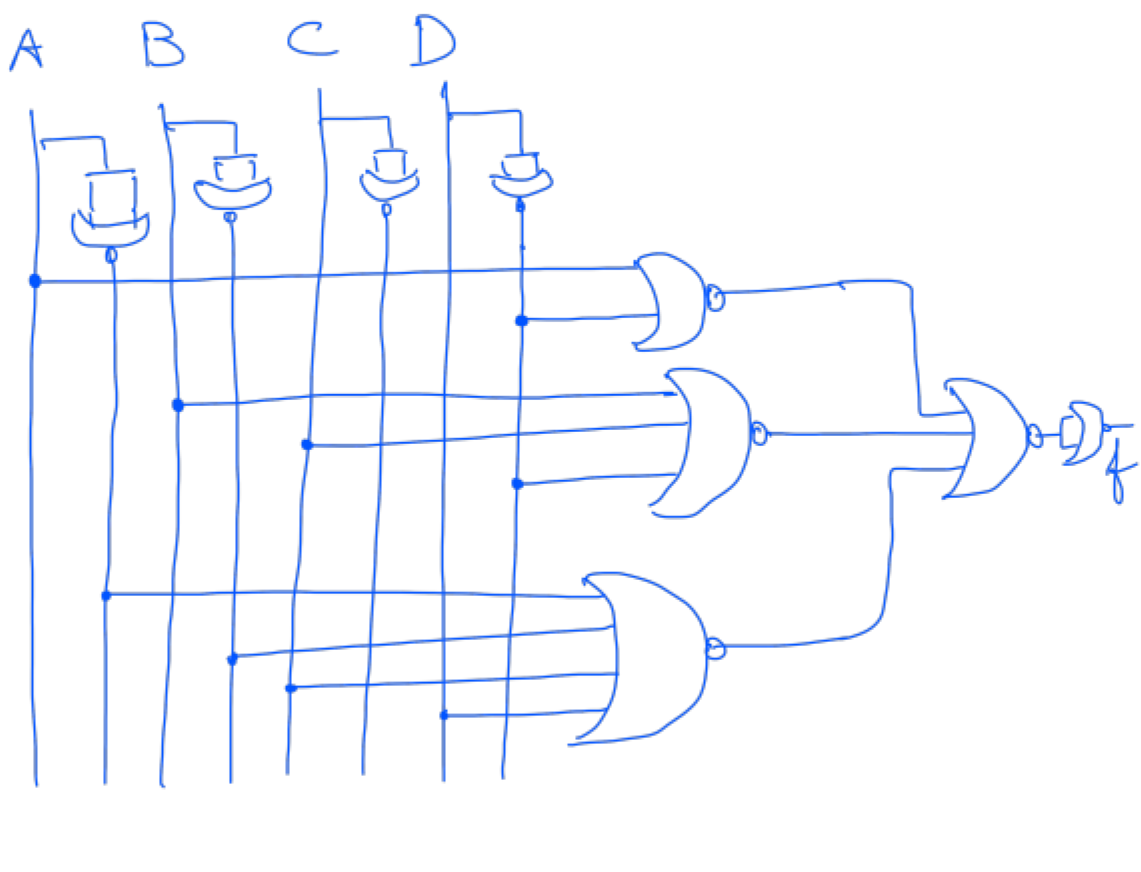
NOR implementation

For NOR implementation, we have to take the complement of the function 'f'

$$\bar{f} = \overline{(\bar{A}D + \bar{B}\bar{C}D + AB\bar{C}\bar{D})}$$

$$= (A+\bar{D})(B+C+\bar{D})(\bar{A}+\bar{B}+C+D)$$

Now implement each term using NOR, then add them using NOR.



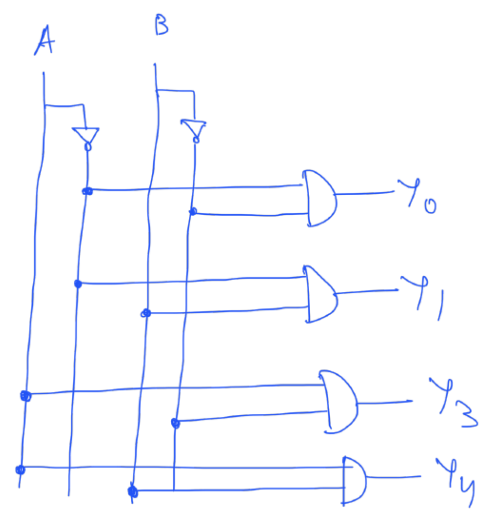
5. Design a 2 × 4 Decoder using logic gates. (*Hint: Assume the decoder is always enabled.*)

Sol.

Truth table for 2x4 decoder

A	B	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$Y_0 = \bar{A}\bar{B}$, $Y_1 = \bar{A}B$, $Y_3 = A\bar{B}$, $Y_4 = AB$



6. Design a 4 to 1 MUX using logic gates. (*A multiplexer, often abbreviated as "MUX", is a digital electronic device that selects one of many input signals and forwards it to a single output line. It operates based on the values of its control inputs, allowing it to route data from multiple sources to a single destination.*)

Sol.

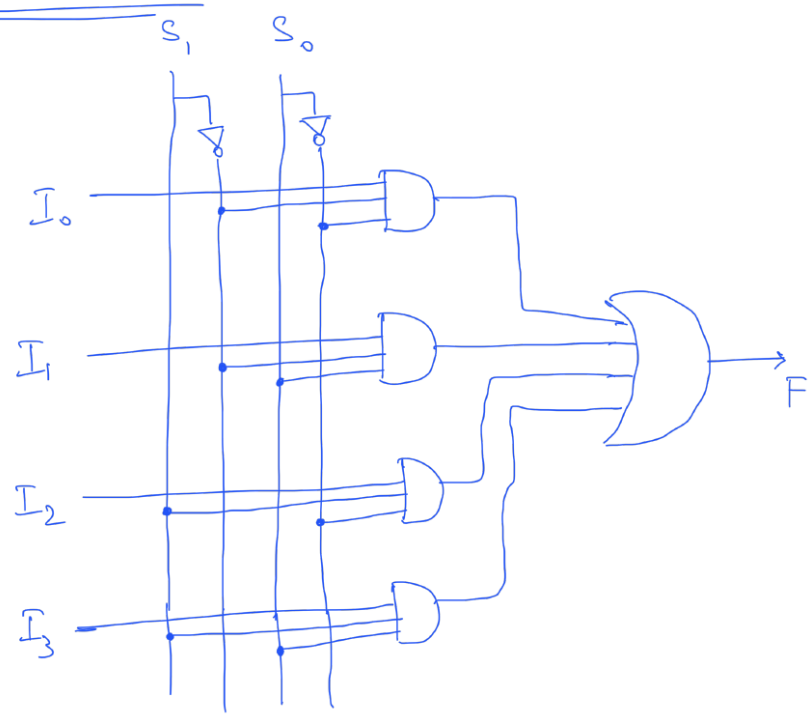
7) 4×1 Mux using logic gates.

Truth table

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Output $F = \bar{S_1}\bar{S_0}I_0 + \bar{S_1}S_0I_1 + S_1\bar{S_0}I_2 + S_1S_0I_3$

Circuit implementation



7. Determine the sum (S = $S_3S_2S_1S_0$), carry (C), and overflow detector (V) for the following cases: (Use the circuit as shown below in fig. 2.)

Sol.

7) a) M = 0 , A = 1010 , B = 1101

M = 0 ⇒ The circuit behaves as an Adder

$$\Rightarrow A + B = \begin{array}{r} 1010 \\ 1101 \\ \hline 1\,0111 \end{array} \begin{array}{l} = A \\ = B \\ \\ \end{array}$$

↑ Carry 'C'

S = 0111

C = 1

V = 0

b) M = 1 , A = 1001 , B = 1100

M = 1 , circuit behaves as subtractor

A − B = A + (2's complement of B)

$$\text{2's complement of } B = \begin{array}{r} 0011 \\ +\quad 1 \\ \hline 0100 \end{array}$$

$$\text{Now: } A - B = \begin{array}{r} 1001 \\ 0100 \\ \hline 1101 \end{array}$$

As A < B , the obtained solution is the 2's complement of B − A

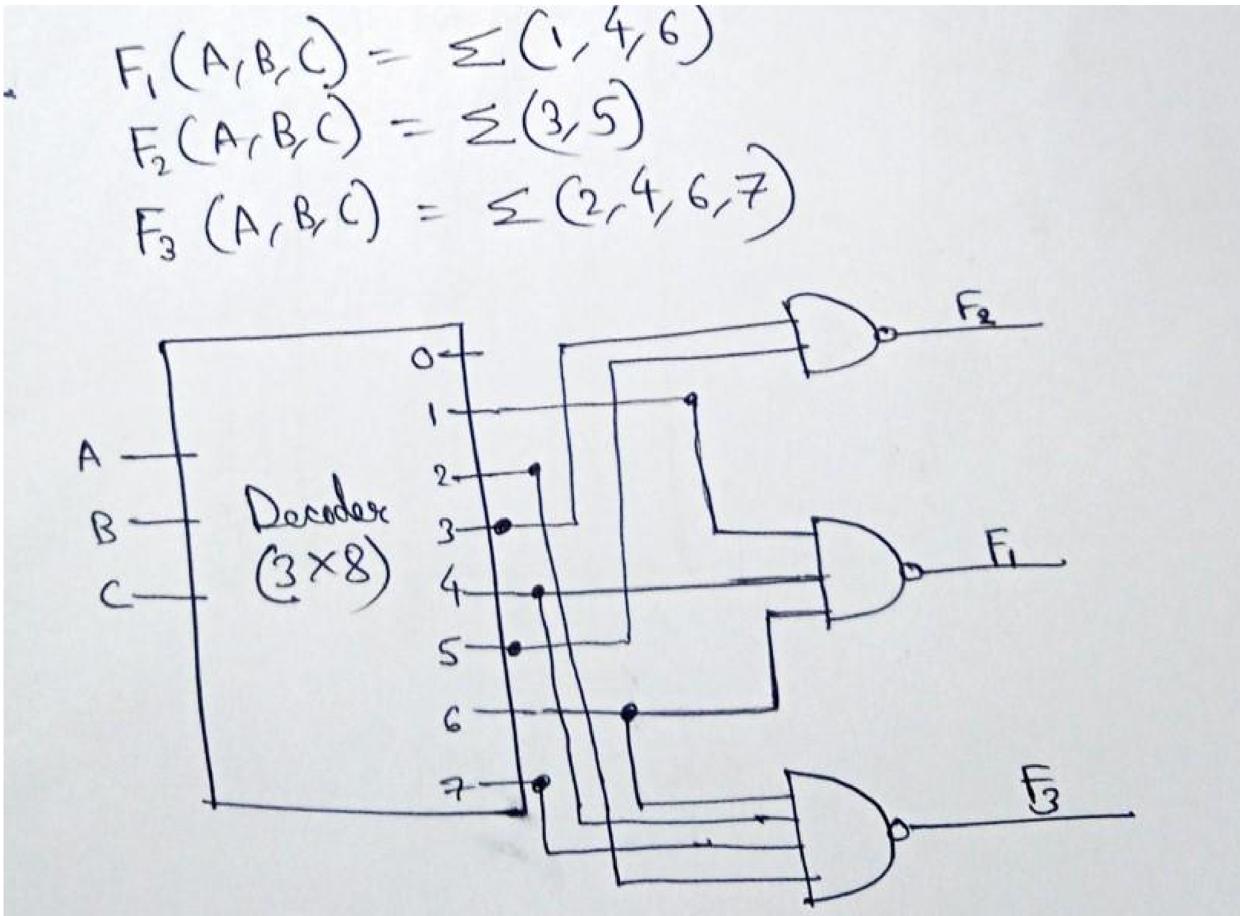
hence the O/P's will be

$$\boxed{\begin{array}{l} S = 1101 \\ V = 0 \\ C = 0 \end{array}}$$

8. A combinational circuit is specified by the following three Boolean functions:

$$F_1(A, B, C) = \Sigma(1, 4, 6)$$
$$F_2(A, B, C) = \Sigma(3, 5)$$
$$F_3(A, B, C) = \Sigma(2, 4, 6, 7)$$

Implement the circuit with a decoder constructed with NAND gates (which means the complements of min-terms are avaible at outputs) and NAND gates connected to the decoder outputs. Use a block diagram for the decoder.

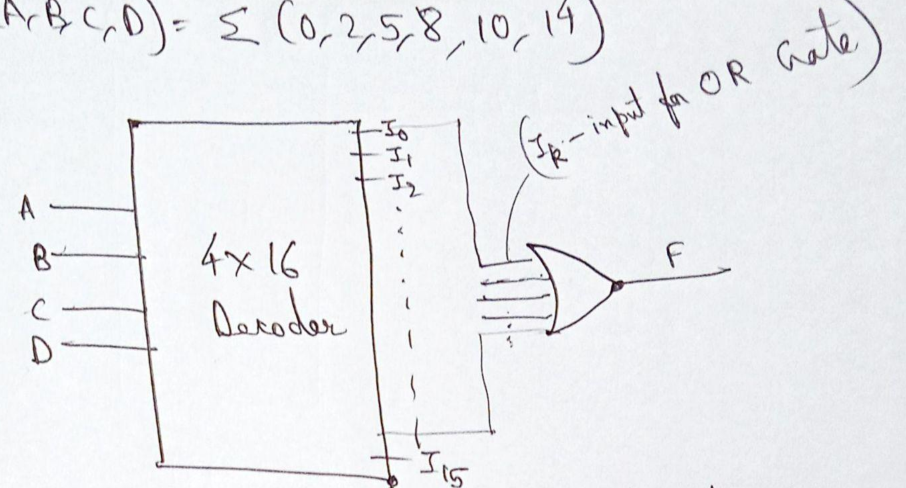


9. Implement the following Boolean function with a decoder of appropriate size.

(a) $F(A, B, C, D) = \Sigma(0, 2, 5, 8, 10, 14)$

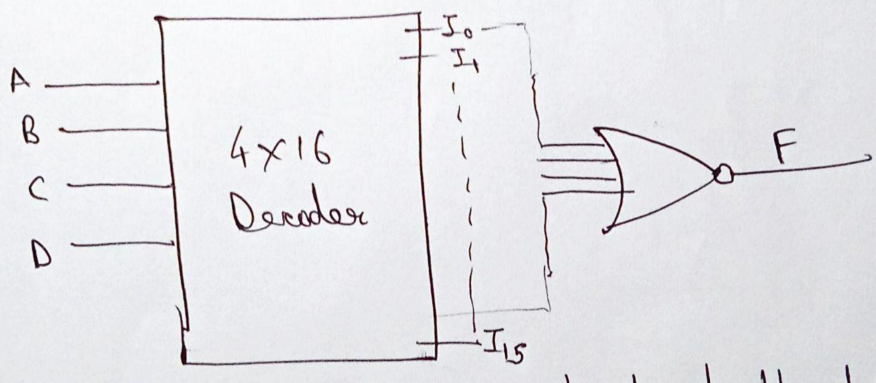
(b) $F(A, B, C, D) = \Pi(2, 6, 11)$

(Ans) (a) $F(A,B,C,D) = \Sigma(0,2,5,8,10,14)$



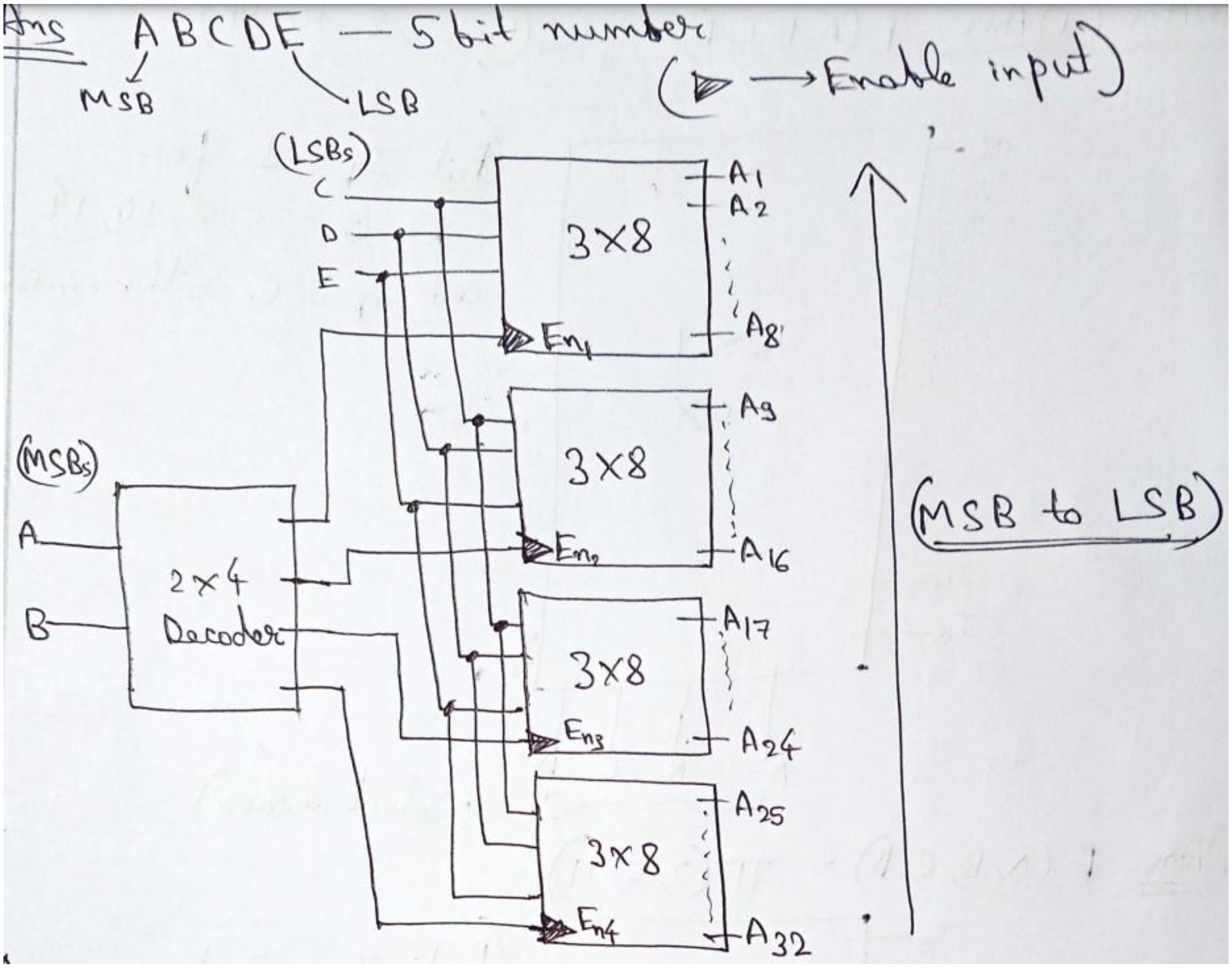
By putting OR Gate at the output of the decoder, and the inputs of OR Gate are "I_k" for $k = 0,2,5,8,10,14$

(b) $F(A,B,C,D) = \Pi(2,6,11)$



By putting NOR Gate at the output of the decoder and the inputs of NOR Gate are :- "I_k" for $k = 2,6,11$

10. Construct a 5-to-32-line decoder with four 3-to-8-line decoders with enable and a 2-to4-line decoder. Use block diagrams for the components.



11. Implement a 4-bit magnitude comparator, which compares two 4-bit binary numbers, using suitable logic gates.

Sol. <u>"Refer to section 4.8 Magnitude comparator (148-150) from the textbook: "Digital Design" by Morris Mano. (5ed)</u>

12. Design a 1-bit comparator (inputs: A and B; which can perform all the 3 operations: greater than, less than and equal to) using a decoder of appropriate size and logic gates. For example, one output of the decoder (say I1) should be equal to 1 only if "greater than" condition is satisfied, keeping the other outputs of the decoder (I0, I2, I3) equal to 0 in this condition. Similarly another output of the decoder (say I2) should be equal to 1 only if "less than" condition is satisfied, keeping the other outputs (I0, I1, I3) equal to 0 in this condition.

Ans We need to design :-

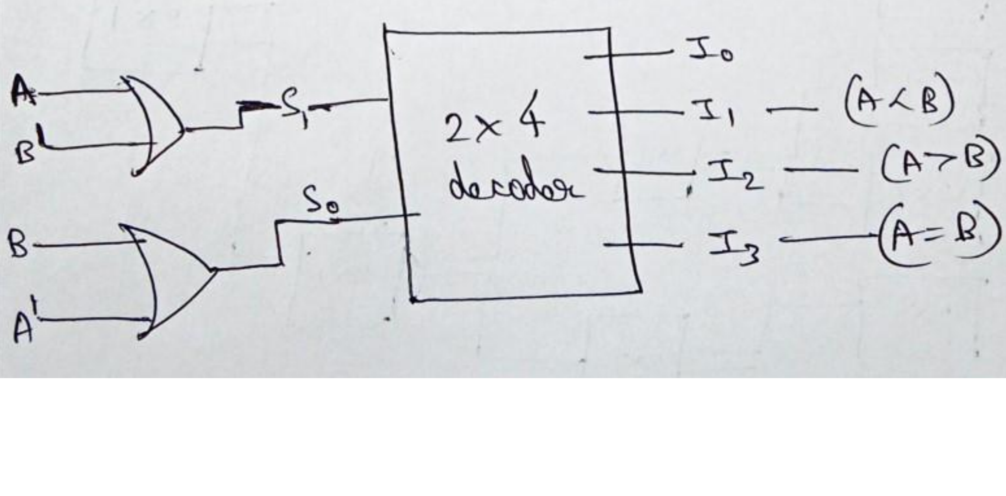
Comparision	Output	
A > B	$I_2 = 1$	$(I_1, I_3 = 0)$
A < B	$I_1 = 1$	$(I_2, I_3 = 0)$
A = B	$I_3 = 1$	$(I_1, I_2 = 0)$

So we make use of 2 × 4 decoder. (S_1, S_0 - select inputs)

	S_1	S_0	I_0	I_1	I_2	I_3	
A < B	0	1	0	1	0	0	
A > B	1	0	0	0	1	0	
A = B	1	1	0	0	0	1	
	0	0	1	0	0	0	— Redundant

covers (A = B = 0 and A = B = 1)

$$S_1 = (A > B) + (A = B)$$
$$= AB' + (A \oplus B)' = AB' + A'B' + AB = \boxed{A + B'}$$
$$S_0 = (A < B) + (A = B) = A'B + A'B' + AB = \boxed{B + A'}$$



Tutorial 4

Practice questions solutions

1. Consider the multiplexer as shown in fig. 3, where, I_0, I_1, I_2, I_3 are four data input lines which can be selected using the two address line combination $A_1A_0 = 00, 01, 10, 11$ respectively and EN is the enable input. Find the expression for 'f' which is implemented by the circuit as shown in fig. 3.

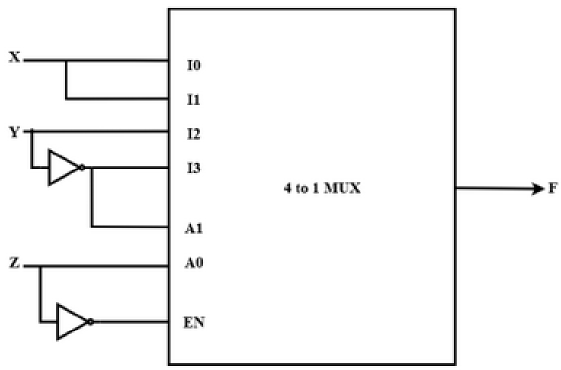


Figure 3: Circuit for Practice Q1

Sol.

O/P of the circuit $f(x,y,z)$

$Z = 0$, then only MUX is enabled.

$\therefore$

A_1	A_0	f	
0	0	x	
0	1	0	$\longrightarrow$ disabled
1	0	y	
1	1	0	$\longrightarrow$ disabled.

$= \bar{A_1}\bar{A_0}\,x + A_1\bar{A_0}\,y$

$= y\bar{z}x + \bar{y}\bar{z}y \quad (\because A_1 = \bar{y}\ \&\ A_0 = z)$

$= \underline{xy\bar{z}}$

2. Design a combinational circuit with two inputs, A, B and one output, Y, using 4×1 MUX and 2×1 MUX. The circuit should give an output of 1 when A and B are both 1 or if A is 0 and B is 1 and 0 otherwise.

Sol.

8) Truth table

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

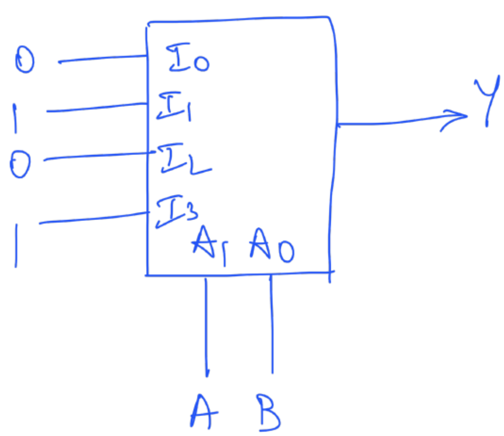
$$Y = \sum m(1,3) = \bar{A}B + AB$$

$$Y = \bar{A}B + AB$$

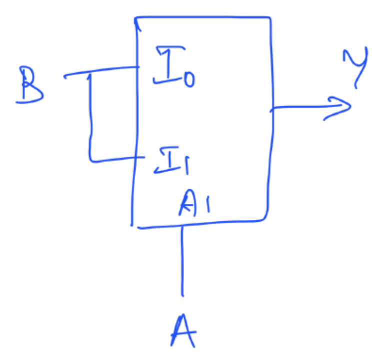
$$= B(A + \bar{A})$$

$$\boxed{Y = B}$$

4x1 MUX



2x1 MUX



3. Design a combinational circuit with two inputs, A, B and one output, Y. The circuit should output 1 if A is greater than or equal to B and 0 otherwise. Draw the resultant circuit using

   a. Using suitable logic gates.
   b. Using 4×1 MUX, and
   c. Using 2×1 MUX.

Sol.

a) Truth table

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

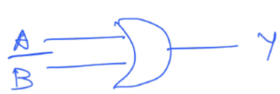
$$Y = \bar{A}\bar{B} + A\bar{B} + AB$$

$$= \bar{A}\bar{B} + A(B+\bar{B})$$

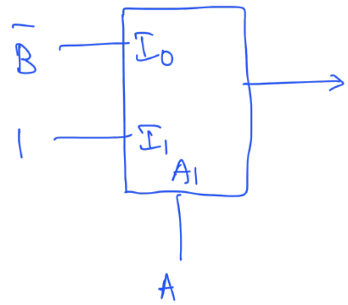
$$= \bar{A}\bar{B} + A$$

$$\boxed{Y = A + \bar{B}}$$

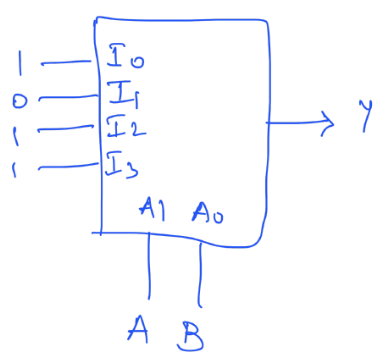
Circuit :



2X1 MUX



4X1 MUX



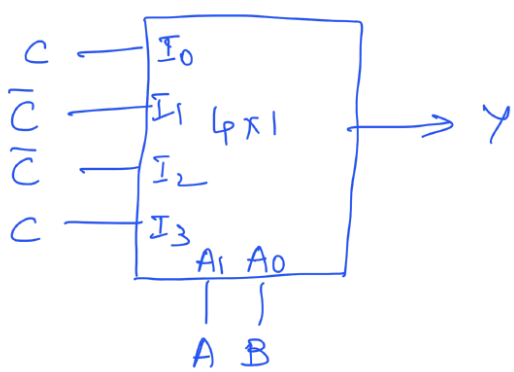
4. Design a combinational circuit with three inputs, A, B and C and one output Y, using 4×1 MUX and a minimum number of 2-input logic gates. The circuit should output a 1 if the number of 1's in the input is odd and 0 otherwise.

Sol.

10) Truth Table

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Implementation using 4x1 MUX :-



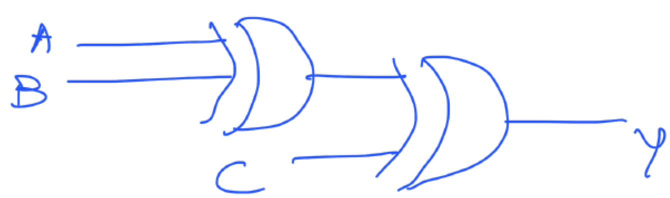
circuit implementation

$$Y = \Sigma m(1, 2, 4, 7)$$

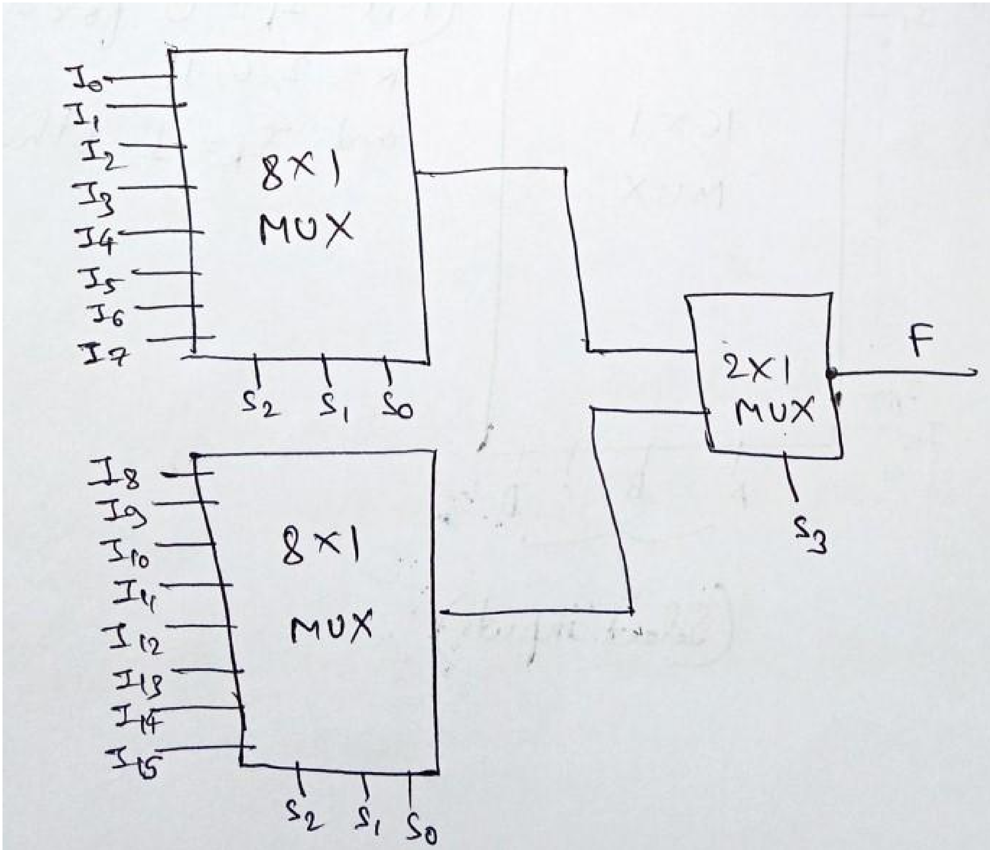
K-Map =

A \ BC	00	01	11	10
0		1		1
1	1		1	

$$Y = A \oplus B \oplus C$$



5. Construct a 16×1 multiplexer with two 8×1 and one 2×1 multiplexer. Use block diagrams.



6. Consider the circuit shown in the figure.

Find the Boolean expression F implemented by this circuit.

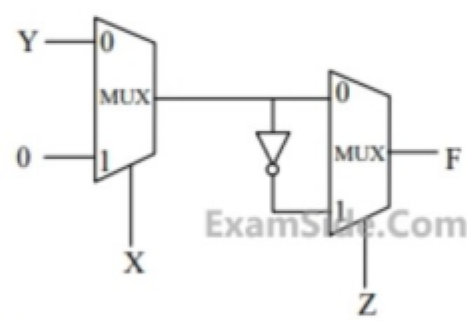
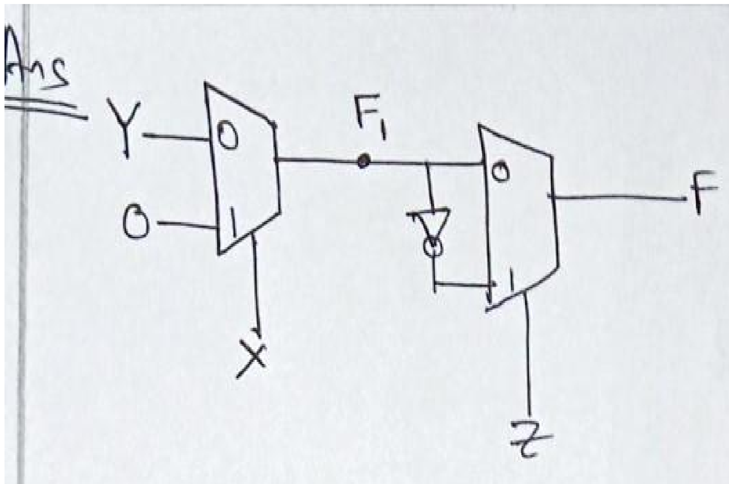


Figure 4: Circuit for Practice Q6

Ans



$F_1 = X'Y + X(0) = X'Y$, $F = Z'F_1 + ZF_1'$

$F = Z'X'Y + Z(X'Y)'$

$= Z'X'Y + Z(X+Y') = \boxed{X'YZ' + Y'Z + XZ}$

7. A 4:1 multiplexer is to be used for generating the output carry of a full adder. A and B are the bits to be added while Cin is the input carry and Cout is the output carry. A and B are to be used as the select bits with A being the more significant select bit. Assign values to

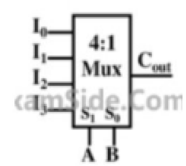
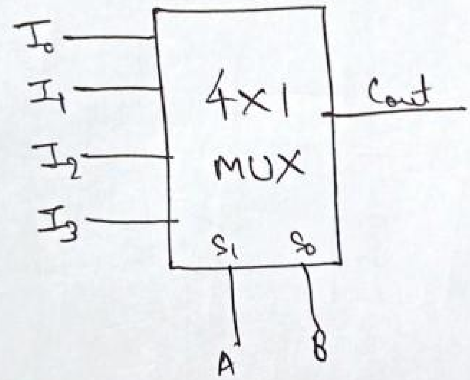


Figure 5: Circuit for Practice Q7

the inputs of the multiplexer such that the output of the multiplexer is equal to C_{out} of full adder.

Ans



$C_{out} = A'B'I_0 + A'BI_1 + AB'I_2 + ABI_3$

By assigning $\boxed{I_0 = 0,\ I_1 = C_{in},\ I_2 = C_{in} \text{ and } I_3 = 1,}$

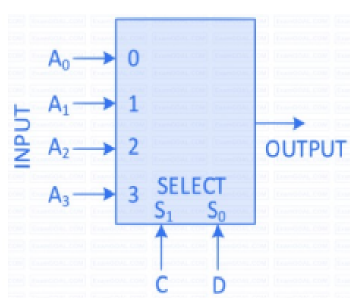
we get $C_{out} = A'B\,C_{in} + AB'\,C_{in} + AB$

$= AB + C_{in}(A \oplus B) \equiv C_{out}$ of full adder.

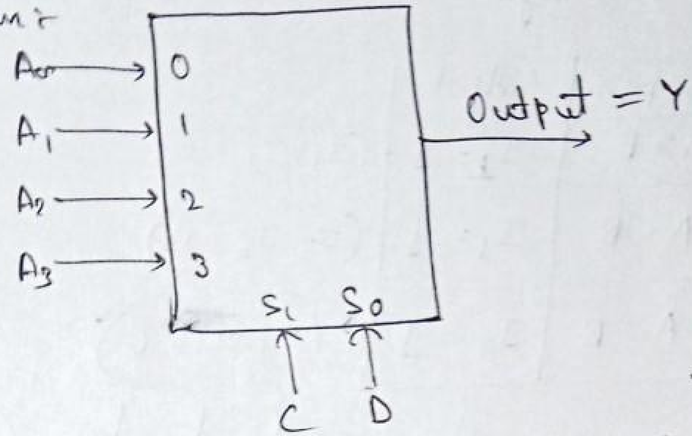
So select $(I_0 = 0,\ I_1 = C_{in},\ I_2 = C_{in},\ I_3 = 1)$

(A – MSB
B – LSB)

8. Consider the 2-bit multiplexer (MUX) shown in the figure 6. For OUTPUT to be the XOR of C and D, the values for A0, A1, A2 and A3 are



Ans Given:-



$$Y = C'D'A_0 + C'DA_1 + CD'A_2 + CDA_3$$

By selecting $\boxed{A_0 = 0,\ A_1 = 1,\ A_2 = 1,\ A_3 = 0}$, we get:-

$$Y = C'D'(0) + C'D(1) + CD'(1) + CD(0)$$

$$\boxed{Y = C'D + CD' = C \oplus D}$$

9. How many 2×1 MUX are required to make 16×1 MUX? Draw the diagram with proper labelling.